

Leveraging psychological data and transparent machine learning for student success prediction in digital learning systems

Musa Adekunle Ayanwale^{a,*}, Olasunkanmi Kehinde^b, Mubarak O Mojinyinola^c,
Chioma Ezech-Kehinde^d

^a Department of Science, Mathematics, and Technology Education, Faculty of Education, University of Pretoria, Groenkloof, 0002, Pretoria, South Africa

^b Department of Psychology, Norfolk State University, United States

^c Department of Psychological and Quantitative Foundations, The University of Iowa, United States

^d Department of Education, Elizabeth City State University, United States

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ABSTRACT

The growth of digital and blended learning has positioned Learning Management Systems (LMS) as core infrastructures for instruction and assessment. Although these platforms generate extensive behavioural data, the transparency and equity of machine learning (ML) models used to predict academic outcomes remain under-examined, particularly in low-resource contexts. This study evaluated the predictive performance, explainability, and fairness of six supervised ML classifiers-Logistic Regression, K-Nearest Neighbours, Support Vector Machine, Random Forest, XGBoost, and Neural Networks using behavioural traces from Thuto (Sakai-based LMS) and survey-derived psychosocial indicators from 850 undergraduates at the National University of Lesotho. Automatically extracted LMS indicators (attempt frequency, discussion engagement, feedback access) were integrated with measures of digital learning self-efficacy, perceived feedback usefulness, trust in AI-supported assessment, and AI ethics awareness. Following preprocessing and a stratified 70–30 train–test split, models were trained and evaluated using accuracy, F1 score, MCC, AUC, confusion matrices, and statistical parity diagnostics. Logistic Regression achieved the highest accuracy (0.714), while Random Forest demonstrated the strongest discrimination (AUC = 0.774) and a favorable balance of classification errors, supporting its suitability for early identification of at-risk learners. Explainability analyses consistently highlighted digital self-efficacy, perceived feedback usefulness, and discussion engagement as dominant predictors across models. Fairness diagnostics indicated comparatively balanced positive prediction rates for linear and tree-based models, whereas neural networks showed instability consistent with overfitting. The findings emphasize that interpretable, fairness-aware modelling can strengthen the trustworthiness and pedagogical value of predictive analytics in digital higher education. Future work should extend fairness evaluation beyond statistical parity and examine real-time implementation through early-warning dashboards and adaptive feedback.

1. Introduction

Predicting student academic performance has become increasingly important in higher education as institutions expand digital, blended, and virtual learning environments. Learning Management Systems (LMS), online assessments, and mobile learning platforms now generate extensive streams of behavioral trace data, creating opportunities to examine learning processes, anticipate academic risk, and design timely interventions. As universities seek to improve retention, reduce

inequities, and enhance instructional decision-making, predictive analytics and machine learning (ML) have emerged as influential tools for translating complex educational data into actionable insights that support student success [1,2].

Within educational data mining (EDM) and learning analytics (LA), a substantial body of research demonstrates that LMS-derived behavioral indicators such as login frequency, assignment submissions, forum participation, resource access, and video interaction metrics can meaningfully predict academic outcomes (Dutt & Ismail, 2018; [3]).

* Corresponding author.

E-mail addresses: ma.ayanwale@up.ac.za (M.A. Ayanwale), ojkehinde@nsu.edu (O. Kehinde), mubarak-mojinyinola@uiowa.edu (M.O. Mojinyinola), ccezeh@ecs.u.edu (C. Ezech-Kehinde).

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Supervised learning techniques including Random Forest, Support Vector Machines, Artificial Neural Networks, Logistic Regression, and Decision Trees have consistently been applied to classify pass–fail outcomes, multi-level grade categories, or continuous performance measures with encouraging levels of predictive accuracy (Al Nagi & Al-Madi, 2020; [4]). Across these studies, engagement behaviors are often interpreted as observable proxies for persistence, self-regulation, and cognitive involvement in digital learning environments [2].

Methodological sophistication within this literature has increased considerably. Traditional classification models validated through cross-validation and standard error metrics remain common (Dutt & Ismail, 2018), while more recent studies integrate clustering, temporal sequence modeling, and multimodal feature extraction to capture behavioral complexity over time (Al Nagi & Al-Madi, 2020). Deep learning architectures have been explored to detect nonlinear patterns within LMS logs and video analytics [5,6].

Comparative research in smart learning environments further indicates that ensemble approaches can enhance predictive robustness when combined with careful preprocessing and feature engineering. For example, Sameephet et al. [7] conducted a comprehensive comparative analysis of multiple ML models to predict English as a Foreign Language (EFL) performance within LMS-based smart learning systems. Their findings demonstrated that ensemble methods, particularly XGBoost and stacking-based architectures, outperformed traditional classifiers in predictive accuracy and stability. The study also emphasized the importance of rigorous preprocessing, feature selection, and validation strategies in optimizing model generalization. Importantly, Sameephet et al. [7] framed predictive analytics within the broader goal of enabling early intervention and personalized learning pathways, thereby positioning performance modelling as a pedagogical tool rather than merely an exercise in algorithmic optimization.

While this line of research establishes the technical superiority of ensemble models in structured educational datasets, important conceptual and contextual limitations remain. First, much of the comparative modelling literature, including Sameephet et al. [7], prioritizes predictive performance while offering comparatively limited discussion of model interpretability and transparency. Black-box ensemble and deep learning architectures may achieve high discrimination accuracy, yet they can obscure the logic underlying predictions. In educational contexts, where algorithmic outputs may influence feedback, formative assessment practices, and academic support decisions, opacity poses practical and ethical concerns [8]. Broader artificial intelligence (AI) scholarship highlights the importance of explainability, accountability, and trust in algorithmic systems ([9]; Molnar, 2020), but these principles are not consistently embedded within LMS-based predictive modelling.

Second, fairness and equity considerations remain underexamined. Although ML models can identify patterns associated with academic risk, they may also reproduce or amplify existing disparities if structural constraints shape digital engagement behaviors [10]. This concern is particularly salient in under-resourced or digitally constrained environments, where differences in internet reliability, device access, data affordability, electricity stability, and digital literacy may influence both LMS activity and predictive outputs [4]. In such contexts, predictive systems risk conflating infrastructural limitations with academic preparedness. Consequently, fairness-aware modelling becomes an essential complement to performance evaluation.

Third, most LMS-based predictive studies originate from well-resourced institutions in North America, Europe, or parts of Asia. Universities in sub-Saharan Africa remain underrepresented in comparative ML research, limiting understanding of how predictive models perform in environments characterized by infrastructural variability. In settings such as the National University of Lesotho, students may rely predominantly on mobile devices, experience fluctuating bandwidth, incur high relative data costs, or share computing resources. These contextual realities shape engagement behaviors and the feature space available for

modelling. Without empirical examination in such settings, conclusions regarding model robustness, fairness, and pedagogical utility remain geographically narrow.

Beyond methodological and contextual gaps, a further limitation concerns conceptual integration. LMS behavioral data are frequently treated as engagement proxies, yet fewer studies explicitly incorporate psychosocial constructs such as digital self-efficacy, trust in AI-supported assessment, feedback literacy, and AI ethics awareness into predictive models. From a social cognitive perspective, self-efficacy influences effort, persistence, and strategic engagement in learning environments. Feedback literacy research emphasizes learners' capacity to interpret and act upon feedback as a determinant of performance [8]. Trust in AI and awareness of ethical implications may shape whether students meaningfully engage with algorithmically mediated assessment systems. Integrating these constructs with observable LMS traces offers a theoretically grounded and socio-educationally meaningful approach to performance prediction.

Explainable AI (XAI) techniques such as SHAP-based and permutation-based importance methods provide mechanisms for clarifying how behavioral and psychosocial features influence classification outcomes ([11]; Molnar, 2020). Embedding such diagnostics within comparative modelling frameworks strengthens interpretability and institutional trust, especially in environments where digital inequities may interact with predictive systems.

Taken together, these gaps highlight the need for predictive models that are not only accurate but also theoretically grounded, interpretable, fairness-aware, and contextually responsive. While studies such as Sameephet et al. [7] demonstrate the technical promise of ensemble methods in smart learning environments, extending this work into digitally constrained contexts requires explicit integration of transparency and equity diagnostics. The present study advances this line of inquiry in three important ways. First, it evaluates ensemble approaches such as Random Forest and XGBoost within a sub-Saharan African LMS environment characterized by infrastructural variability. Second, it systematically incorporates explainable AI diagnostics to clarify the contribution of psychosocial and engagement-related features. Third, it integrates fairness analysis to assess whether predictive outcomes remain equitable in a digitally constrained context. By embedding ensemble performance evaluation within a framework of transparency and equity, this study moves LMS-based predictive modelling beyond accuracy-centric paradigms.

1.1. Purpose of the study

Against this backdrop, the present study evaluates the predictive performance, interpretability, and fairness of multiple supervised machine learning classifiers using integrated LMS engagement indicators and psychosocial variables. Specifically, the study aims to:

- (a) determine which ML models most effectively classify student pass–fail outcomes within a digitally mediated assessment environment;
- (b) identify the behavioral and psychological features that most strongly influence predictions using explainable AI techniques; and
- (c) assess group-level fairness and transparency within an under-represented sub-Saharan African higher education context.

By integrating predictive accuracy with explainability, psychosocial theory, and fairness diagnostics, this study advances the development of trustworthy, pedagogically meaningful AI applications for digital assessment systems. It contributes to the learning analytics literature by situating ML performance prediction within a broader socio-technical framework that recognizes infrastructural constraints, learner dispositions, and ethical accountability as integral components of responsible educational AI.

2. Methodology

2.1. Research design and context

This study adopted a cross-sectional analytical design integrating LMS behavioral trace data with survey-based psychosocial and access-related variables to model student academic outcomes. The design reflects contemporary learning analytics practice, where digital trace data are combined with theoretically grounded learner attributes to enhance predictive and explanatory power [12,13].

The study was conducted within Thuto, the official Learning Management System (LMS) of the National University of Lesotho (NUL). Thuto is built on the Sakai open-source framework and supports quizzes, discussion forums, assignment submission, and feedback modules. These functionalities generate timestamped interaction logs, enabling extraction of traceable behavioral indicators for predictive modelling.

The context is particularly relevant for examining AI-supported prediction in digitally constrained environments. In addition to variability in internet reliability, students at NUL may experience fluctuating bandwidth, reliance on mobile devices, shared access to computing resources, and differential digital literacy levels. Such contextual conditions can shape LMS engagement patterns and may influence predictive outputs. Investigating model performance and fairness in this setting therefore contributes to expanding the geographic and infrastructural diversity of learning analytics research, which has been dominated by well-resourced institutional contexts.

Undergraduate students who participated in at least one LMS-mediated assessment activity during the semester were eligible for inclusion, ensuring that all participants generated meaningful behavioral traces. A stratified random sampling procedure was employed to ensure representation across gender, field of study, and year level. Stratification supports subgroup comparison and fairness diagnostics. The final analytic sample consisted of $N = 850$ students, providing sufficient variability for classification modelling while maintaining representativeness across demographic categories.

2.2. Instrumentation and variable operationalization

This study employed a multimodal instrumentation strategy that integrates automatically logged behavioral trace data from the Thuto Learning Management System (LMS) with validated psychosocial survey measures and structural access indicators. The rationale for this integration is grounded in contemporary learning analytics scholarship, which emphasizes that digital performance prediction is strengthened when observable engagement behaviors are modelled alongside theoretically informed learner dispositions [12,13]. Rather than treating LMS interaction logs as isolated predictors, the present design conceptualizes student performance as the product of an interaction between behavioral engagement, motivational-cognitive readiness, and infrastructural access conditions. This approach enhances both predictive robustness and interpretive coherence, particularly in digitally evolving and resource-constrained educational environments.

Behavioral variables were extracted directly from timestamped Thuto server logs and operationalized as indicators of digital engagement and self-regulated learning behaviors (Li et al., 2021). Importantly, to ensure temporal validity and prevent data leakage, all behavioral predictors were computed within a defined observation window that concluded prior to the final summative assessment deadline. This design ensures that the features used for modelling strictly precede the outcome variable, thereby maintaining causal plausibility and methodological integrity.

Three primary LMS behavioral indicators were included. First, *Attempt Frequency* was measured as a continuous variable ranging from 0 to 10, representing the number of attempts made on LMS-mediated quizzes or formative assessments during the observation window. In digital learning theory, repeated attempts can reflect persistence,

mastery orientation, or the use of an iterative learning strategy. However, excessive attempts may also signal uncertainty or difficulty, making it an analytically informative yet theoretically ambiguous indicator.

Second, *Discussion Engagement Level* was operationalized as an ordinal variable coded from 0 (None) to 4 (Very High), reflecting the intensity of participation in LMS discussion forums. Unlike simple login metrics, forum participation captures dialogic engagement, peer interaction, and collaborative knowledge construction. In online learning research, such dialogic behaviors are frequently interpreted as stronger indicators of cognitive engagement and persistence than passive access metrics.

Third, *Feedback Response Behavior* was coded as a binary indicator (0 = No, 1 = Yes) reflecting whether a student accessed instructor-provided feedback within 72 h of release and at least seven days before the summative assessment closing date. Feedback events occurring after the summative deadline were excluded to prevent temporal leakage. This operationalization aligns with feedback literacy theory, which emphasizes timely feedback uptake as a precursor to performance improvement. By restricting the measurement window, the study ensures that the predictor reflects engagement with formative feedback rather than post-outcome review behavior. Together, these LMS-derived indicators provide ecologically valid measures of observable learning activity, avoiding reliance on self-report and reducing common-method bias.

In addition to behavioral traces, psychosocial constructs (see full item sets in the appendix) were measured through a structured questionnaire administered on a five-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree). These constructs extend predictive modelling beyond observable behavior by incorporating theoretically grounded learner dispositions that may shape how students interact with digital systems.

Self-Efficacy in Digital Learning (SEDL) was adapted from Zimmerman's [14] social cognitive framework and extended through applications in online learning contexts [15]. Self-efficacy reflects students' perceived capability to navigate digital platforms, solve technical problems, and manage online academic tasks. From a theoretical perspective, higher digital self-efficacy is expected to influence persistence, strategic engagement, and proactive feedback use within LMS environments.

Feedback Usefulness Rating (FUR) was adapted from Nicol and Macfarlane-Dick's [16] feedback model. Rather than measuring mere receipt of feedback, this construct captures students' perception of clarity, actionability, and developmental value. In digital assessment contexts, perceived usefulness may mediate the extent to which feedback is acted upon, thereby influencing subsequent performance trajectories.

AI Ethics Awareness (AIEA), adapted from Floridi and Cowls [17], captures students' understanding of ethical principles governing AI systems in education, including bias awareness and considerations for data use. While not traditionally included in performance prediction models, its inclusion reflects the study's broader commitment to fairness and transparency in AI-mediated assessment environments.

Trust in AI Assessment (TAIA), adapted from Long and Magerko [18], reflects students' confidence in AI-generated grading or feedback systems. Trust may influence behavioral compliance, feedback uptake, and willingness to rely on automated evaluation mechanisms. As AI integration increases, these socio-technical trust variables may meaningfully shape engagement patterns.

All psychosocial constructs demonstrated satisfactory internal consistency (Cronbach's $\alpha \geq 0.70$) and were aggregated into composite scores in accordance with recommended psychometric procedures [19]. By integrating these constructs, the study operationalizes performance prediction within a theoretically anchored framework rather than relying solely on behavioral proxies.

To account for structural digital inequities that may influence engagement patterns, two technology access variables were included.

Internet Reliability was measured on a five-point self-rating scale (1 = Very Poor to 5 = Excellent), reflecting perceived stability and consistency of connectivity. In digitally constrained contexts, unstable internet access may directly limit LMS interaction frequency, forum participation, and timely feedback review. Including this variable enables examination of whether predictive models inadvertently encode infrastructural disadvantage.

Prior AI Literacy was coded as a binary indicator (0 = No, 1 = Yes), representing self-reported familiarity with AI tools used in education. Exposure to AI systems may shape trust, ethical awareness, and interaction patterns in response to AI-mediated feedback.

Lastly, the outcome variable *Academic Performance* was operationalized as a binary classification derived from institutional gradebook records, where scores below 50 were coded as 0 (Fail) and scores of 50 and above as 1 (Pass). Binary classification aligns with the design of an early-warning system, where the primary objective is to identify students at risk of failing and support timely intervention. Although this dichotomization simplifies performance variability, it enhances interpretability and supports fairness diagnostics at the group level.

Collectively, this instrumentation strategy constructs a holistic feature space integrating behavioral traces, psychosocial dispositions, infrastructural conditions, and objective performance outcomes (see Fig. 1). Such integration aligns with contemporary calls for transparent, theory-informed, and equity-sensitive AI applications in educational assessment [20], while maintaining methodological rigor through temporal alignment and leakage prevention.

2.3. Data collection procedures

Data collection followed a structured, ethically governed, and temporally controlled protocol designed to ensure methodological rigor, reproducibility, and protection against data leakage. The procedure was aligned with established standards in learning analytics and educational data mining [12,13], while incorporating safeguards necessary for fairness-aware predictive modelling. Prior to data collection, ethical clearance was obtained from the university's institutional review board. All participants provided digital informed consent in the learning management system. The consent form explicitly authorized the linkage of LMS behavioral trace data with survey responses and institutional gradebook records for research purposes. Participation was voluntary, and students retained the right to withdraw without penalty at any stage prior to anonymized dataset construction.

2.4. Participant recruitment and consent

Eligible undergraduate students were identified through active enrollment in at least one LMS-mediated assessment during the semester. Invitations were disseminated through official Thuto announcement channels to ensure equitable access to participation. The invitation outlined the study's purpose, confidentiality safeguards, data-handling

procedures, and terms of voluntary participation. Only students who digitally affirmed consent were granted access to the survey instrument. This consent-first architecture ensured that no behavioral trace data were processed for students who did not explicitly authorize participation, thereby maintaining compliance with ethical data governance principles.

2.5. Survey administration

The psychosocial survey was administered within Thuto through a secure embedded survey link. All items required mandatory responses to minimize missingness and prevent partial submissions. Response time metrics were monitored to identify inattentive or careless responding, and extreme completion times were flagged for review. The survey captured demographic characteristics and theoretically grounded psychosocial constructs, including self-efficacy in digital learning, feedback usefulness, AI ethics awareness, and trust in AI assessment. These constructs were measured before students knew the final assessment outcomes, thereby preserving the temporal integrity between the predictors and the outcome.

2.6. LMS trace data extraction and temporal integrity

Following survey completion, LMS behavioral variables were extracted using timestamp-matched student identifiers. Extraction scripts accessed server log records that documented quiz attempts, forum participation, and feedback access events. To prevent temporal leakage, all behavioral predictors were computed within a predefined observation window that concluded prior to the final summative assessment deadline. Specifically:

- Attempt frequency included only attempts logged before the final assessment submission date.
- Discussion engagement was calculated based on forum participation during the instructional period, excluding post-assessment activity.
- Feedback response behavior was coded only if feedback was accessed within 72 h of release and at least seven days before the final assessment deadline.

Events occurring after grade release or assessment closure were excluded. This time-bound feature construction ensured that predictive models were trained exclusively on information available prior to outcome realization, thereby preserving causal plausibility and preventing future information from contaminating model training.

2.7. Data cleaning, matching, and quality assurance

Survey responses and LMS trace data were merged using anonymized student identifiers. Records from non-consenting participants and incomplete cases were removed prior to analysis. Range validation checks were conducted on all variables to detect invalid values, and multicollinearity diagnostics were performed to assess model stability. Outlier screening was conducted using standardized residual thresholds for continuous variables. Feature distributions were examined to assess skewness and potential imbalance effects. The final analytic dataset consisted of $N = 850$ students and included behavioral, psychosocial, access-related, and performance variables suitable for classification modelling and fairness diagnostics.

2.8. Model preparation and feature engineering

Prior to model training, preprocessing procedures were implemented to ensure algorithm compatibility and prevent data leakage. Continuous variables were standardized using z-score normalization based exclusively on the training subset to prevent information bleed into the test set. Nominal categorical variables were one-hot encoded. Ordinal

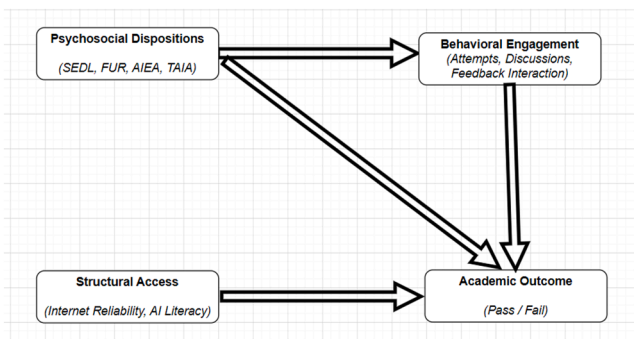


Fig. 1. Conceptual framework.

variables were retained as ordered categorical inputs where appropriate. To preserve the class distribution of pass and fail cases, a stratified 70–30 train–test split was implemented. Stratification ensured proportional representation of lower-frequency fail cases in both subsets, thereby reducing majority-class bias and supporting fairness diagnostics [12].

The 70 % training partition provided adequate sample size for parameter estimation and hyperparameter tuning, while the 30 % hold-out test set enabled robust generalization assessment. All preprocessing transformations were learned on the training set and subsequently applied to the test set to maintain strict separation.

2.9. Machine learning workflow and model training

We implemented a structured machine learning workflow (see Fig. 2) to predict students' assessment outcomes utilizing behavioral, attitudinal, and access-related variables derived from the Thuto LMS. The modelling procedures were conducted using JASP 0.18, which provides an integrated environment for supervised classification, evaluation metrics, hyperparameter tuning, and explainable AI diagnostics. Following preprocessing, the dataset was separated into features X and the target label y . A 70–30 stratified train–test split was applied to preserve the distribution of pass/fail classes, ensuring stability and validity in model evaluation.

2.10. Data analysis and model specification

To evaluate predictive performance, interpretability, and robustness across modelling paradigms, six supervised machine learning algorithms were implemented and compared. These models were selected to represent complementary methodological families frequently employed in educational data mining research, ranging from interpretable linear classifiers to nonlinear ensemble and neural architectures. This comparative design enables evaluation not only of predictive accuracy but also of transparency, generalization stability, and suitability for fairness-aware educational deployment.

The selection of these six algorithms was also informed by prior comparative research in smart LMS environments. Sameephet et al. [7] demonstrated that ensemble methods, particularly XGBoost, frequently outperform single classifiers when predicting academic performance in structured educational datasets. However, their work primarily emphasized predictive strength and model optimization. Building on this evidence, the present study integrates both ensemble and non-ensemble approaches to systematically examine trade-offs between predictive power, interpretability, and distributive fairness. This expanded comparative lens is especially important in a sub-Saharan African higher education context, where infrastructural variability and

digital access disparities may interact with algorithmic decision processes. Consequently, model evaluation is framed not only in terms of performance superiority but also in terms of transparency, equity sensitivity, and contextual suitability for responsible educational deployment.

Let the feature space be defined as:

$$X = \{x_1, x_2, x_3, \dots, x_p\}$$

and the binary outcome variable as:

$$y \in \{0, 1\}$$

where 0 = Fail and 1 = Pass.

The modelling objective was to learn a function:

$$f: X \rightarrow \{0, 1\}$$

such that:

$$\hat{y} = f(X)$$

minimizes classification error while maintaining interpretability and stability.

2.10.1. Logistic regression model

Logistic Regression (LR) was used as a transparent baseline model for benchmarking interpretability. The binary dependent variable Y_i represented students' performance (1 = Pass, 0 = Fail), modelled as a log-odds function of the predictors:

$$\text{logit}(P_i) = \ln\left(\frac{P_i}{1-P_i}\right) = \beta_0 + \sum_{j=1}^9 \beta_j X_{ij} + \varepsilon_i$$

where:

P_i = probability that student i passes (score ≥ 50),

X_{ij} = j -th predictor for student i (e.g., Attempt frequency, discussion engagement level, etc.)

β_j = regression coefficient estimating feature contribution.

ε_i = random error term.

Model coefficients were estimated through maximum likelihood estimation (MLE), and statistical significance was assessed at $p < .05$. Odds ratios ($OR = e^{\beta}$) quantified effect magnitude for each predictor. The LR model also provided a benchmark for calibration and interpretability comparisons against more complex algorithms.

2.10.2. Random forest model

The Random Forest (RF) algorithm [21] was selected as the primary ensemble model given its robustness and capacity to handle heterogeneous data types. It operates by constructing an ensemble of T uncorrelated decision trees, each trained on a bootstrap sample of the dataset.

For a given observation x , the predicted class probability is:

$$\hat{P}(Y=1|x) = \frac{1}{T} \sum_{t=1}^T h_t(x)$$

Where $h_t(x)$ denotes the predicted probability from tree t . The final classification is determined by majority voting across all trees. Also, feature importance in RF was computed through Gini impurity reduction, where the importance of variable j is:

$$I_j = \frac{1}{T} \sum_{t=1}^T \sum_{n \in N_t(j)} p(n) \Delta i(n)$$

Where, $p(n)$ is the proportion of samples reaching node n in tree t , and $\Delta i(n)$ represents the Gini decrease due to splitting on variable j . This metric ranks predictors by their average contribution to reducing classification uncertainty.

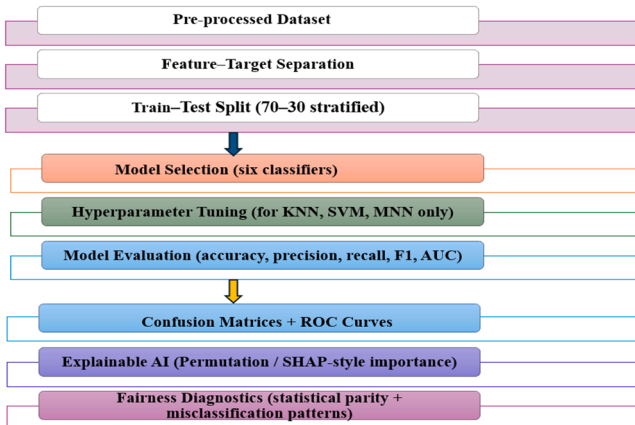


Fig. 2. Machine learning workflow.

2.10.3. Multilayer perceptron (Neural network model)

To capture potentially nonlinear relationships among behavioral, attitudinal, and access-related features, we implemented a Multilayer Perceptron (MLP), configured using four hidden layers as illustrated in Fig. 5. The architecture consists of an input layer receiving the feature vector:

$$X = \{x_1, x_2, x_3, \dots, x_n\},$$

four hidden layers (32–24–16–8 neurons), and a single-neuron output layer predicting class membership (Pass = 1 / Fail = 0). Each neuron performs an affine transformation followed by an activation function, thus enabling the network to approximate nonlinear decision boundaries.

For a given hidden layer l , the forward propagation is expressed as:

$$h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)}),$$

Where $h^{(l)} = X$

$W^{(l)}$ is the weight matrix, $b^{(l)}$ is the bias vector, and $\sigma(\cdot)$ represents the nonlinear activation (ReLU in the hidden layers). The output layer applies to a sigmoid function to generate a probability score:

$$\hat{y} = \phi(W^{(L)}h^{(L-1)} + b^{(L)})$$

With

$$\phi(z) = \frac{1}{1 + e^{-z}}, \quad \hat{y} \in (0, 1).$$

A binary cross-entropy loss function was used during training:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Optimization used the Adam optimizer (learning rate = 0.001) over 200 epochs with early stopping to mitigate overfitting. Dropout regularization $p = 0.2$ was applied after each hidden layer to enhance generalization. This configuration enabled the neural network to learn complex interaction patterns among LMS engagement, attitudinal readiness, and infrastructural access. However, performance was inferior to tree-based models, as indicated by wider train-validation gaps and greater overfitting sensitivity, consistent with patterns observed in low-resource student data where sample variability and feature noise are prominent.

2.10.4. K-Nearest Neighbor classifier (KNN)

We applied the K-Nearest Neighbor (KNN) algorithm, which operates on the principle that students with similar behavioral and attitudinal profiles are likely to achieve comparable performance outcomes. The algorithm classifies a new student by examining the k most similar students in the dataset and assigning the predominant class (pass or fail) among them as the prediction (Cover & Hart, 1967).

Within the LMS context, similarity was computed using Euclidean distance across features such as attempt frequency, discussion engagement, feedback access, and self-efficacy scores. The stratified training-validation process helped identify the optimal number of neighbors ($k = 7$), achieving the best balance between sensitivity and specificity.

2.10.5. Support vector machine (SVM)

We adopted the Support Vector Machine (SVM) algorithm to classify students' assessment outcomes using behavioral, attitudinal, and access-related predictors. The SVM aims to find the optimal decision boundary (known as the maximum-margin hyperplane) that separates the pass and fail classes in the feature space. Its objective is to maximize the margin between the two classes, thereby improving generalization and reducing classification error (Cortes & Vapnik, 1995). We used a cost parameter ($C = 0.01$), which controls the trade-off between classification accuracy and margin width. Lower values prioritize generalization over perfect

classification, which is important for reducing overfitting.

2.10.6. Gradient Boosting / XGBoost (XGB)

Extreme Gradient Boosting (XGBoost) has emerged as a robust tool for predicting student assessment outcomes, particularly in the context of structured educational data. Its efficacy stems from its ability to integrate a diverse array of variables, including behavioral, attitudinal, and access-related factors. In essence, XGBoost constructs a series of decision trees, where each subsequent tree learns from the missteps of its predecessor. This sequential learning approach not only enhances the model's predictive power but also allows it to elucidate intricate relationships among various variables, such as the frequency of attempts, levels of self-efficacy, engagement, and responsiveness to feedback. In practice, XGBoost's suitability for this study is underscored by several key features. It adeptly manages both categorical and continuous data, which is particularly beneficial in educational research where data is often diverse and multifaceted. Additionally, the model incorporates built-in regularization techniques designed to mitigate the risk of overfitting, a common concern in machine learning applications, especially in smaller datasets. Furthermore, XGBoost generates clear rankings of feature importance, which align well with the principles of explainable AI and can facilitate deeper insights into the factors influencing student performance.

2.11. Model performance evaluation

Model performance was evaluated using ten-fold cross-validation within the training data, followed by assessment on the independent hold-out test set. Metrics were averaged across folds to ensure stability. The following evaluation indices were computed (see Appendix for formulae): Accuracy, Precision, Recall, F1 Score, Matthews Correlation Coefficient (MCC), and Area Under the ROC Curve (AUC), where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. AUC was used to evaluate threshold-independent discrimination capability. MCC was included because it provides a balanced measure under class imbalance. Confusion matrices were analyzed to examine error asymmetries, particularly false negatives (missed at-risk students) and false positives (misclassified failing students). One-vs-rest ROC curves were used to visualize model separability across thresholds. Comparative evaluation considered predictive accuracy, calibration behavior, computational efficiency, interpretability, and fairness stability to ensure that conclusions were not based solely on accuracy metrics.

3. Results

3.1. Demographic profile of respondents

A total of 850 undergraduate students from the National University of Lesotho participated in the study. The sample was gender-balanced, with 48.9 % male and 51.1 % female, allowing for meaningful subgroup comparisons in fairness diagnostics. The age distribution mirrored typical undergraduate enrollment patterns, with the majority of students aged 21–23 years (44.6 %), followed by 18–20 years (37.3 %) and 24–26 years (18.1 %). Representation across year levels was also adequate, with Year 2 students making up the largest proportion (36.2 %), followed by Year 3 (30.4 %), Year 1 (21.8 %), and Year 4 (11.6 %). Importantly, internet reliability, a variable sensitive to fairness, exhibited significant variability. While 68.7 % reported good or excellent connectivity, 31.3 % reported fair to very poor connectivity. In a digitally mediated assessment context, this variability poses a potential structural constraint that could influence engagement patterns and predictive outputs. Additionally, 57.5 % reported high prior AI literacy, indicating diversity in technological familiarity. This variation enhances the robustness of fairness diagnostics and ensures that model behavior can be evaluated across both digitally advantaged and constrained

subgroups (see Table 1).

3.2. Measurement model evaluation

Prior to conducting predictive modelling, the psychometric adequacy of all survey-based constructs was assessed to ensure that the attitudinal variables used as machine learning features met standard reliability and validity requirements. A Confirmatory Factor Analysis (CFA) was conducted to examine the internal structure of the latent constructs: Self-Efficacy in Digital Learning (SEDL), Feedback Usefulness Rating (FUR), AI Ethics Awareness (AIEA), and Trust in AI Assessment (TAIA). The resulting measurement model demonstrated strong internal consistency and robust construct validity (see Table 2).

All constructs showed satisfactory reliability, with Cronbach's alpha values ranging from 0.83 to 0.88 and Composite Reliability (CR) values between 0.86 and 0.90, confirming stable unidimensionality and consistency across scale items. Convergent validity was well-supported, as Average Variance Extracted (AVE) values ranged from 0.55 to 0.64, exceeding the recommended threshold of 0.50 [19]. Factor loadings for all items were statistically significant ($p < .001$), with standardized coefficients ranging from 0.67 to 0.86, indicating that the items captured substantial variance in their respective underlying constructs. Additionally, we assessed the discriminant validity using the Fornell–Larcker criterion [22].

For each construct, the square root of AVE was greater than its inter-construct correlations, demonstrating clear conceptual separation among the latent variables. No evidence of multicollinearity or cross-loading was detected, as all variance inflation factor (VIF) values remained below 3.3 [23], further affirming the integrity of the measurement model. The overall model fit indices indicated an excellent fit to the data ($\chi^2/df = 2.31$, CFI = 0.94, TLI = 0.93, RMSEA = 0.049, with a 90 % CI of 0.044–0.056). These values exceed the recommended benchmarks for well-fitting measurement models, confirming that the constructs were psychometrically sound and suitable for use in the subsequent machine learning classification procedures.

3.3. Machine learning classification

Six supervised learning algorithms were trained and evaluated to predict students' assessment outcomes using Thuto LMS behavioral indicators and attitudinal variables. The models included Logistic Regression (LR), K-Nearest Neighbors (KNN), Random Forest (RF), Support Vector Machine (SVM), Extreme Gradient Boosting (XGBoost), and a Multilayer Neural Network (MNN). Each model was trained using

Table 1
Demographic characteristics of respondents ($N = 850$).

Variable	Category	Frequency (n)	Percentage(%)
Gender	Male	416	48.9
	Female	434	51.1
Age Group (years)	18–20	317	37.3
	21–23	379	44.6
	24–26	154	18.1
	27–30	100	11.9
Year of Study	Year 1	185	21.8
	Year 2	308	36.2
	Year 3	258	30.4
	Year 4	99	11.6
Field of Study	Education	288	33.9
	Science and Technology	224	26.4
	Social Sciences	178	20.9
	Humanities	160	18.8
Internet Reliability	Very Poor	38	4.5
	Poor	79	9.3
	Fair	149	17.5
	Good	361	42.5
Prior AI Literacy	Excellent	223	26.2
	Low	361	42.5
	High	489	57.5

Table 2

Measurement model reliability and validity indices.

Construct	A	CR	AVE	Standardized Loading Range (λ)
Self-Efficacy in Digital Learning	0.88	0.90	0.64	0.72 – 0.86
Feedback Usefulness Rating	0.84	0.87	0.58	0.70 – 0.82
AI Ethics Awareness	0.83	0.86	0.55	0.67 – 0.80
Trust in AI Assessment	0.85	0.88	0.60	0.71 – 0.84

Note. Model Fit: $\chi^2/df = 2.31$, CFI = 0.94, TLI = 0.93, RMSEA = 0.049.

All factor loadings are significant at $p < .001$. CR ≥ 0.70 , AVE ≥ 0.50 .

standardized features and evaluated on an independent test set, with performance assessed using accuracy, precision, recall, F1 score, Matthews Correlation Coefficient (MCC), Area Under Curve (AUC), the Receiver Operating Characteristic Curves (ROC), and fairness indicators such as statistical parity. Table 3 presents an integrated comparison of all models, while the corresponding confusion matrices are displayed in Fig. 1A.

3.3.1. Logistic regression

Logistic Regression provided a strong baseline classifier, achieving a test accuracy of 0.714 and an AUC of 0.713. The model demonstrated balanced predictive capability across both classes, with precision values of 0.719 (Pass) and 0.709 (Fail), and recall values of 0.690 and 0.736, respectively. An MCC of 0.427 indicated a moderate positive association between predicted and observed classifications, confirming stable generalization. Moreover, the confusion matrix (see Fig. 1A) revealed relatively low false-positive and false-negative rates, supporting the suitability of Logistic Regression for transparent and interpretable prediction in educational settings.

3.3.2. K-Nearest Neighbors

The KNN model ($k = 7$) achieved an accuracy of 0.624 and an AUC of 0.647. Although moderate in performance, KNN exhibited symmetry in precision and recall (0.636/0.610) and produced balanced classification outcomes across classes. However, KNN showed a higher false-positive rate, particularly misclassifying failing students as passing, which is undesirable in early-warning academic systems (see Fig. 1A).

3.3.2.1. Model tuning and hyperparameter optimization. To identify optimal configurations and minimize overfitting, we evaluated hyperparameter tuning curves for the KNN, multilayer neural network, and SVM models. Fig. 3 illustrates how training and validation accuracy changed as key parameters were adjusted, providing insight into the stability and generalization of each model. Across algorithms, training accuracy consistently exceeded validation accuracy, indicating varying degrees of overfitting. However, the magnitude and pattern of divergence differed across models, underscoring the need for model-specific tuning. For the KNN classifier, validation accuracy increased steadily with larger neighborhood sizes and peaked at $k = 7$, after which gains stabilized. Small k values produced inflated training accuracy, demonstrating high sensitivity to noise. The curve, therefore, supports $k = 7$ as the most balanced and generalizable configuration. Meanwhile, the neural network displayed progressive improvement in both training and validation accuracy across epochs; however, the widening gap between the curves indicates that the model gradually overfitted the training data. The plateauing of validation accuracy below the training trajectory aligns with the weaker predictive performance observed in the results. For the SVM, accuracy was highly sensitive to the cost parameter. Validation accuracy fluctuated at higher cost values but stabilized at $C \approx 0.01$, which delivered the best generalization. Increasing the cost boosted training accuracy but reduced validation accuracy, reflecting overfitting through an overly rigid decision boundary. The low-cost configuration, therefore, provided the most appropriate balance.

Table 3
Model performance metrics.

Metric	LR	KNN (k = 7)	RF	XGBoost	SVM	MNN
Accuracy	0.714	0.624	0.682	0.665	0.612	0.300
Precision (PPV)	0.719 (Pass), 0.709 (Fail)	0.636 / 0.610	0.653 / 0.720	0.693 / 0.642	0.639 / 0.586	0.254 / 0.333
Recall (TPR)	0.690 / 0.736	0.636 / 0.610	0.747 / 0.621	0.605 / 0.726	0.596 / 0.630	0.214 / 0.384
False Positive Rate (FPR)	0.264 / 0.310	0.390 / 0.364	0.379 / 0.253	0.274 / 0.395	0.370 / 0.404	0.616 / 0.786
False Discovery Rate (FDR)	0.281 / 0.291	0.364 / 0.390	0.347 / 0.280	0.307 / 0.358	0.361 / 0.414	0.746 / 0.667
F1 Score	0.704 / 0.722 / 0.714	0.636 / 0.610 / 0.624	0.697 / 0.667 / 0.681	0.646 / 0.682 / 0.664	0.616 / 0.607 / 0.612	0.232 / 0.357 / 0.295
MCC	0.427	0.246	0.370	0.333	0.225	-0.408
AUC	0.713	0.647	0.774	0.699	0.613	0.686
NPV	0.709 / 0.719	0.610 / 0.636	0.720 / 0.653	0.642 / 0.693	0.586 / 0.639	0.333 / 0.254
TNR (Specificity)	0.736 / 0.690	0.610 / 0.636	0.621 / 0.747	0.726 / 0.605	0.630 / 0.596	0.384 / 0.214
FNR	0.310 / 0.264	0.364 / 0.390	0.253 / 0.379	0.395 / 0.274	0.404 / 0.370	0.786 / 0.616
FOR	0.291 / 0.281	0.390 / 0.364	0.280 / 0.347	0.358 / 0.307	0.414 / 0.361	0.667 / 0.746
Threat Score	0.813 / 0.848 / 0.831	0.583 / 0.521 / 0.552	0.713 / 0.720 / 0.716	0.650 / 0.670 / 0.660	0.552 / 0.500 / 0.526	0.105 / 0.178 / 0.142
Statistical Parity	0.475 / 0.525 / 1.000	0.518 / 0.482 / 1.000	0.559 / 0.441 / 1.000	0.441 / 0.559 / 1.000	0.488 / 0.512 / 1.000	0.418 / 0.582 / 1.000

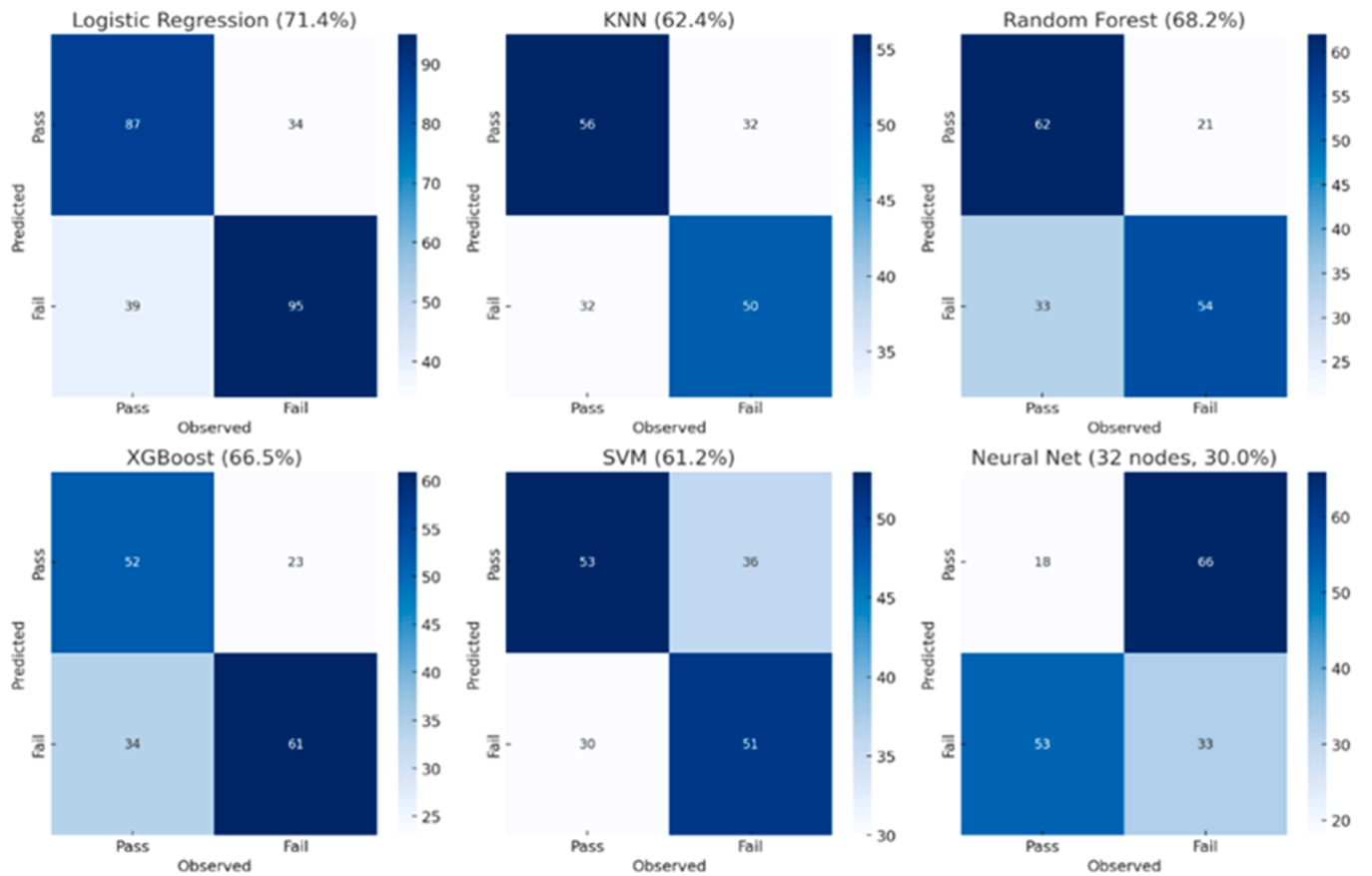


Fig. 3. Confusion Matrix for all the models.

Nevertheless, hyperparameter optimization was not performed for Logistic Regression, Random Forest, and XGBoost. These models exhibit strong algorithmic stability and reliable performance. Additional tuning would yield marginal gains at substantial computational cost and was therefore unnecessary.

3.3.3. Multilayer neural network

Two neural network architectures were tested. The 4-layer network with 16 hidden nodes achieved a test accuracy of 0.641 (AUC = 0.614), but suffered from sensitivity to class imbalance, leading to high false

positive and false negative rates. The larger network, with 32 nodes, performed even more poorly (accuracy = 0.300), exhibiting negative MCC values, which indicate overfitting and unstable generalization (see Table 3).

3.3.4. Random forest

Random Forest emerged as the best-performing classifier, achieving a test accuracy of 0.682 and the highest AUC value across models (0.774). The model also demonstrated strong recall for the Pass class (0.747) and balanced overall performance (F1 = 0.681, MCC = 0.370) as

presented in Table 4. The RF confusion matrix showed lower misclassification of failing students compared to KNN and the neural networks, indicating superior suitability for early detection of at-risk students (see Fig. 1A).

3.3.5. Extreme gradient boosting

XGBoost performed competitively, with an accuracy of 0.665 and an AUC of 0.699. The model displayed a strong balance between sensitivity and specificity (recall = 0.605/0.726) and demonstrated robust handling of nonlinear relationships. Although XGBoost underperformed relative to Random Forest, its stable MCC (0.333) confirmed reliable predictive behavior.

3.3.6. Support vector machine

The SVM classifier achieved modest performance (accuracy = 0.612, AUC = 0.613) (see Table 4). Precision and recall values were symmetrical across classes; however, the model struggled to detect failing students, resulting in elevated false negative rates (see Fig. 1A). Although theoretically powerful, SVM did not outperform tree-based models in this educational context.

Fig. 1A reveals clear differences in how each model classified the pass and fail classes. As expected, the tree-based models proved to be the most stable. Both Random Forest and XGBoost achieved the best balance between false positives and false negatives, demonstrating consistent sensitivity and specificity. Random Forest misclassified 33 failing students as passing and 21 passing students as failing, while XGBoost showed a similarly balanced pattern. Logistic Regression also performed reliably, producing symmetrical errors (39 false positives and 34 false negatives), which indicates stable class sensitivity and robust baseline performance.

By contrast, KNN misclassified a high number of failing students as passing (32 false positives), suggesting that it tended to default to the majority class when uncertainty arose. SVM showed an opposite misclassification trend, producing a substantially high number of false negatives (36), indicating that it struggled to detect borderline pass cases. The neural network models performed poorly across both classes, generating large misclassification errors in both directions. This instability confirms that the network overfitted during training and failed to generalize to unseen data.

Additionally, across all models, each ROC curve (see Fig. 4)

illustrates the trade-off between sensitivity (true positive rate) and specificity (true negative rate) at varying decision thresholds. Random Forest and XGBoost displayed the strongest class separability, as reflected in larger ROC areas and consistent diagnostic reliability. Logistic Regression also demonstrated robust discriminatory capacity, while KNN and SVM produced moderate separability. Neural Networks showed inconsistent performance, confirming limited generalization and susceptibility to overfitting. These ROC patterns align with the outcomes of the confusion matrix and reinforce the suitability of tree-based and linear models for the early identification of at-risk learners in digital learning environments.

3.4. Feature importance and model explainability

To enhance the interpretability of ML models and align with fairness-driven assessment practices, we implemented a structured explainable AI (XAI) framework to examine how individual predictors contributed to model performance. Rather than relying solely on internal feature weights or impurity measures, we applied permutation-based mean dropout loss across classifiers. This model-agnostic procedure estimates the marginal contribution of each feature by randomly permuting its values and observing the resulting decline in predictive performance (operationalized as change in AUC). A larger increase in prediction error following permutation indicates greater feature importance. Permutation importance is conceptually aligned with Shapley Additive exPlanations (SHAP), in that both approaches quantify the marginal contribution of a feature relative to model output [11]. However, whereas SHAP decomposes individual predictions using cooperative game theory, permutation importance evaluates global feature contributions by measuring performance degradation when information is disrupted (Molnar, 2020). This distinction is important for interdisciplinary audiences: SHAP provides local explanation at the instance level, while permutation importance provides global explanation at the model level. In this study, the global approach was prioritized because the research objective focused on identifying stable predictors of student success across modelling paradigms rather than explaining individual student-level predictions.

Table 4 reports mean dropout loss values computed over 50 permutations for each classifier. Despite differences in algorithmic architecture, a consistent structure emerged across models. Three predictors, self-efficacy in digital learning, feedback usefulness rating, and discussion engagement level, demonstrated the highest and most stable marginal contributions. In the Logistic Regression model, self-efficacy (mean dropout loss = 0.385), feedback usefulness (0.363), and discussion engagement (0.357) produced the largest decreases in predictive accuracy when permuted, and all associated regression coefficients were statistically significant. Because Logistic Regression is inherently interpretable through odds ratios, the permutation results serve as a robustness check, confirming that both parametric coefficients and global perturbation analyses converge on the same explanatory structure. This convergence strengthens confidence in the theoretical validity of the feature space.

The similarity-based and margin-based classifiers (KNN and SVM) exhibited comparable patterns. In KNN, discussion engagement (0.286) and trust in AI assessment (0.283) ranked highest. This suggests that in proximity-based classification, students' active participation and attitudinal orientation toward AI-mediated assessment contribute meaningfully to similarity clustering and outcome discrimination. SVM likewise ranked discussion engagement and self-efficacy among the top predictors, indicating that even under maximum-margin optimization, engagement-related features dominate decision boundaries. Importantly, the consistency of these predictors across linear, similarity-based, and margin-based algorithms indicates that their importance is not an artifact of a particular modelling paradigm.

A contrasting explanatory profile emerged in the neural network models. Permutation results indicated greater sensitivity to

Table 4
Feature importance across ML models.

Feature	LR	KNN	RF	SVM	XGBoost	MNN
Self-efficacy in digital learning	0.385	0.274	0.012 / 0.012	0.344	36.107 / 0.355	0.667
Discussion engagement level	0.357	0.286	0.008 / 0.014	0.359	18.516 / 0.323	–
Feedback response behavior	0.340	0.255	0.009 / 0.011	0.322	0.000 / 0.259	0.682
Feedback usefulness rating	0.363	0.277	0.011 / 0.003	0.321	17.772 / 0.302	0.637
Attempt frequency	0.334	0.249	0.003 / 0.001	0.329	5.659 / 0.269	0.681
AI ethics awareness	0.330	0.279	0.006 / 0.005	0.318	13.267 / 0.295	0.683
Trust in AI assessment	0.331	0.283	0.002 / 0.001	0.313	5.568 / 0.263	0.680
Internet reliability	0.316	0.253	0.004 / –0.001	0.290	3.111 / 0.265	0.709
Prior AI literacy	0.318	0.255	0.002 / 0.00036	0.282	0.000 / 0.259	–

Note. Mean dropout loss is defined as $(1 - \text{AUC})$ and is based on 50 permutations. Random Forest reports a mean decrease in accuracy and an increase in node purity; XGBoost includes a relative influence ranking.

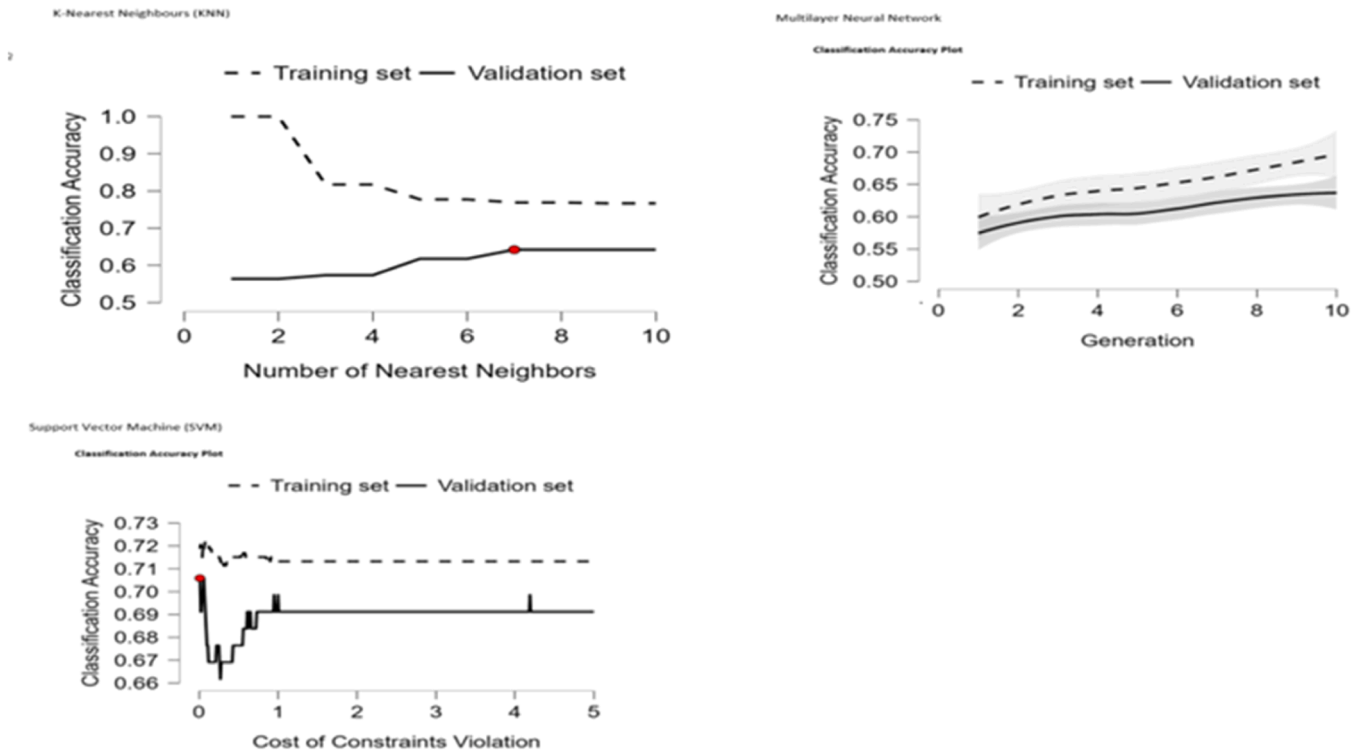


Fig. 4. Hyperparameter optimization.

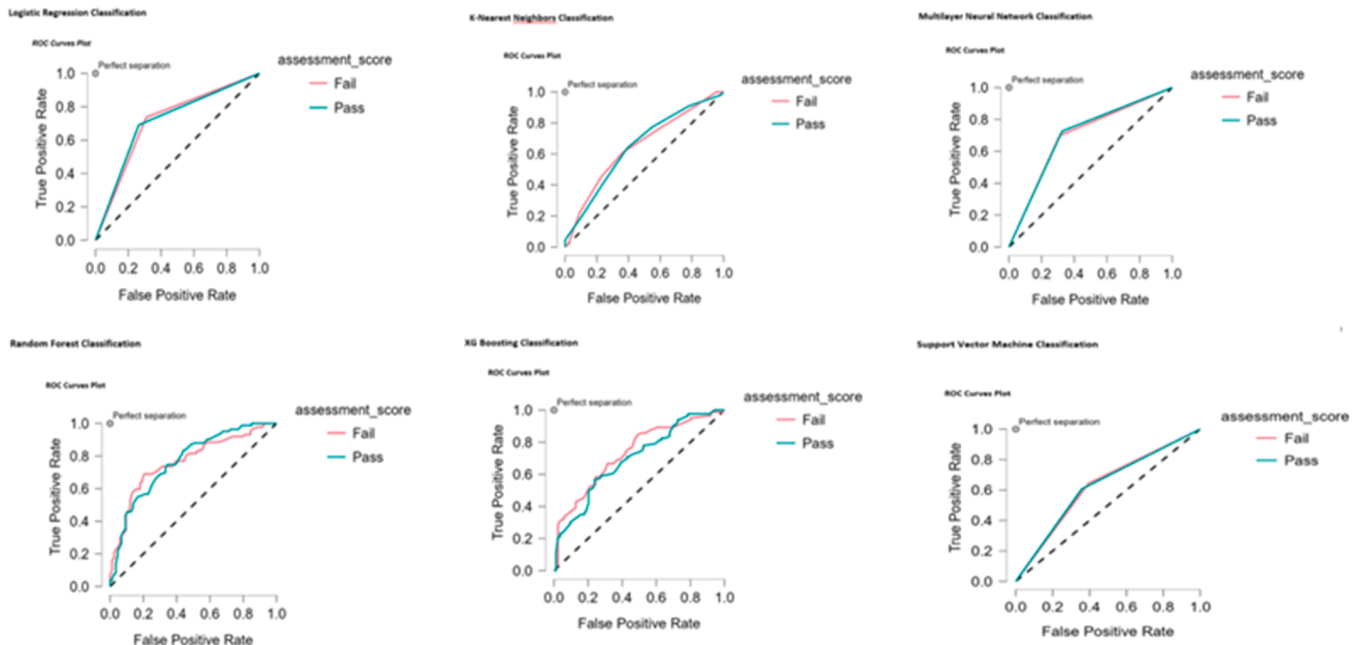


Fig. 5. ROC curves for the models.

infrastructural variables, particularly internet reliability and AI ethics awareness. The disproportionately high dropout loss associated with these features suggests that neural architectures may internalize contextual variability as predictive signal. When considered alongside the model's reduced test accuracy and evidence of overfitting, this pattern implies that the neural network may have learned spurious associations linked to structural digital inequality rather than stable learning-relevant features. From a fairness perspective, this reinforces concerns that complex models deployed in resource-variable

environments may encode contextual disadvantage into predictive outputs.

Tree-based ensemble methods demonstrated both strong predictive performance and theoretically coherent feature attribution. Random Forest importance, evaluated using mean decrease in accuracy and Gini impurity reduction, highlighted discussion engagement, self-efficacy, and feedback responsiveness as dominant variables. XGBoost similarly assigned the highest relative influence to self-efficacy (36.1 %), followed by discussion engagement (18.5 %) and feedback usefulness (17.8 %).

The agreement between permutation-based importance and tree-specific impurity metrics indicates cross-method stability in feature ranking. Essentially, the convergence of global permutation importance, parametric coefficients, and ensemble impurity measures reveals a robust behavioral–motivational axis underlying student performance. Engagement in dialogic activity, confidence in digital learning, and active interpretation of feedback collectively form the core predictive structure. The stability of this feature hierarchy across modelling families enhances interpretability and supports theoretical alignment with self-regulated learning and feedback literacy frameworks [8].

3.5. Fairness diagnostic

To assess distributive equity, statistical parity was examined across models (Table 3). Statistical parity evaluates whether predicted positive outcomes (Pass = 1) are proportionally distributed across groups, independent of ground truth labels. Values near 0.50 indicate balanced distribution, while extreme values suggest skewness. Across models, statistical parity ranged from 0.418 to 0.559, indicating generally acceptable distributional balance. Logistic Regression, Random Forest, and XGBoost demonstrated the most stable parity values, suggesting no systematic favoring of either predicted class.

Neural network models showed the greatest imbalance (0.418/0.582), consistent with their instability and higher error asymmetry. Importantly, examination of confusion matrices revealed no evidence of severe class-based distortion in tree-based or linear models. Although statistical parity does not capture conditional error disparities (as in equalized odds), its inclusion is appropriate in an early-warning educational context where equitable allocation of positive classification is central. When interpreted alongside class-specific error rates, results indicate that ensemble and linear models maintained both predictive strength and acceptable distributive fairness within this low-resource LMS environment.

4. Discussion

This study examined the predictive performance, interpretability, and fairness of six supervised machine learning algorithms for classifying student academic outcomes using LMS engagement indicators and psychosocial variables within a sub-Saharan African university context. By integrating ensemble modeling, explainable AI diagnostics, and fairness analysis, the study moves beyond accuracy-centric approaches and situates predictive analytics within a broader socio-technical and pedagogical framework. The findings corroborate established trends in educational data mining and extend prior work by demonstrating how algorithmic behavior interacts with learner dispositions and infrastructural variability in digitally uneven environments.

In line with comparative research in smart learning systems, tree-based ensemble methods demonstrated the strongest discrimination capacity. Random Forest achieved the highest AUC (0.774), followed by XGBoost (0.699), while Logistic Regression provided a transparent and competitively performing baseline (0.713). These results are consistent with evidence that ensemble methods often outperform linear and distance-based classifiers in structured educational datasets characterized by nonlinear engagement patterns (Dutt & Ismail, 2018; [4]). They also align with Sameephet et al. [7], who reported the predictive superiority of ensemble approaches such as XGBoost in LMS-based environments. However, whereas prior comparative work primarily emphasized gains in predictive accuracy, the present study embeds ensemble evaluation within interpretability and fairness diagnostics, thereby extending performance benchmarking toward responsible educational deployment.

The findings indicate that ensemble effectiveness derives more from structural compatibility with heterogeneous educational data than from algorithmic complexity. LMS engagement traces and psychosocial indicators frequently interact in nonlinear and conditional ways, and tree-

based partitioning appears well-suited to capture such relationships without requiring the large datasets typically necessary for neural network generalization. In contrast, the neural network architectures exhibited clear overfitting, unstable misclassification patterns, and reduced generalization. This mirrors prior observations that deep learning models require substantial data volumes and refined feature engineering to perform robustly in educational contexts [2]. In digitally constrained settings, where engagement data may be uneven and infrastructural variability introduces noise, ensemble and linear models may therefore offer greater operational reliability and interpretability.

Across modeling paradigms, self-efficacy in digital learning, discussion engagement, and feedback usefulness consistently emerged as the most influential predictors. Their stability across classifiers reinforces their theoretical and practical significance. From a social cognitive perspective, self-efficacy underpins persistence, effort regulation, and strategic engagement, making its prominence in predictive modeling theoretically coherent. The importance of feedback usefulness aligns with feedback literacy scholarship, which emphasizes learners' capacity to interpret and act upon feedback as central to achievement [8]. The combined influence of perceived feedback quality and actual feedback engagement suggests that effective uptake, rather than mere availability of feedback, drives performance. Similarly, the strong predictive role of discussion engagement corroborates learning analytics research indicating that active participation predicts academic outcomes more reliably than passive metrics such as logins or page views [3]. Together, these predictors form a behavioral–motivational axis that integrates observable activity with learner dispositions, addressing a gap in prior LMS-based modeling studies that often analyze trace data independently of psychosocial constructs.

A noteworthy distinction emerged in the neural network models, which assigned disproportionate importance to infrastructural variables such as internet reliability. This pattern suggests that complex architectures may amplify contextual signals linked to structural digital inequalities rather than isolating learning-relevant behaviors. Such sensitivity resonates with broader concerns in AI scholarship that models trained in low-resource environments may inadvertently encode infrastructural disadvantage into predictive outputs [1,24]. In contrast, ensemble and linear models demonstrated greater stability in feature attribution, prioritizing engagement and motivational variables over access-related constraints. This has methodological implications: predictive systems deployed in digitally variable contexts must distinguish between academic engagement and infrastructural limitations. Models that are overly responsive to connectivity variability risk conflating resource constraints with learner capability.

Fairness diagnostics revealed generally acceptable statistical parity across models, with Logistic Regression, Random Forest, and XGBoost exhibiting the most balanced predicted outcome distributions. Although statistical parity does not capture conditional error disparities such as equalized odds, its interpretation alongside confusion matrices and class-specific error rates indicates that predictive strength did not coincide with pronounced group-level imbalance. Neural network models showed greater instability in predicted outcome distribution, reinforcing concerns regarding their suitability for contexts where equitable decision-making is essential. These findings align with emerging scholarship suggesting that interpretable and ensemble-based approaches may be better suited for fairness-aware educational applications [25]. In digitally heterogeneous environments, fairness requires not only distributive balance but also stability across infrastructural subgroups.

The results also carry operational implications for LMS-based early warning systems. The prominence of self-efficacy, feedback usefulness, and discussion engagement suggests that predictive dashboards should extend beyond frequency-based activity counts. Integrating indicators of feedback uptake, perceived feedback clarity, and dialogic participation may provide more pedagogically meaningful signals of academic risk. Moreover, in contexts where a substantial proportion of students report

unreliable internet access, model deployment must account for infrastructural variability. Lightweight ensemble models such as Random Forest and XGBoost offer strong predictive performance without excessive computational demands, making them viable in resource-constrained institutional settings.

By situating ensemble modeling within a socio-technical framework that incorporates learner dispositions, contextual constraints, explainability, and fairness, the study advances LMS-based predictive research beyond purely technical comparisons. It demonstrates that predictive accuracy, interpretability, and distributive balance can be jointly evaluated without sacrificing methodological rigor. The findings further suggest that model complexity alone does not guarantee superior performance; rather, alignment between algorithmic structure and data characteristics is more decisive. Tree-based ensemble methods balance flexibility and robustness, while logistic regression remains a valuable benchmark for transparency and calibration.

5. Implications for research and practice

The findings offer important implications for advancing both research and practice in AI-supported assessment and learning analytics. First, the strong and stable performance of Logistic Regression, Random Forest, and XGBoost underscores the practical value of models that balance predictive strength with interpretability. These algorithms not only achieved reliable classification accuracy but also maintained relatively balanced error distributions across subgroups. For practitioners, this highlights that model selection should prioritize transparency and stability rather than complexity alone. Integrating explainable AI tools into LMS dashboards can enhance instructional trust by clarifying how specific engagement and attitudinal variables shape predictions, thereby supporting informed pedagogical decision-making.

Second, the consistent prominence of self-efficacy, discussion engagement, and feedback usefulness identifies clear leverage points for instructional intervention. These are modifiable aspects of the learning process rather than fixed background characteristics. Strengthening students' confidence in digital learning, cultivating meaningful peer interaction, and promoting effective feedback engagement may directly influence academic outcomes. Institutions can operationalize these insights by embedding structured feedback cycles, dialogic discussion protocols, and targeted self-efficacy development initiatives within digital learning strategies. Early-warning systems, therefore, should not function solely as risk detectors but as diagnostic tools that guide actionable instructional responses.

Third, the results reinforce the importance of addressing digital inequities alongside predictive modelling. The neural network models' heightened sensitivity to infrastructural variables illustrates how complex algorithms may inadvertently encode structural disadvantages. This finding emphasizes that fairness-aware evaluation must accompany predictive deployment. For researchers, this entails embedding fairness diagnostics into routine model assessment. For institutional leaders, it underscores that AI innovation should be coupled with investments in connectivity, device access, and digital literacy development. Predictive systems cannot be ethically separated from the infrastructural realities within which they operate.

Finally, the study demonstrates the feasibility of implementing ensemble-based early-warning models in resource-constrained environments. Random Forest and XGBoost exhibited balanced error profiles and computational efficiency, making them suitable for formative feedback and academic advising contexts. However, responsible implementation requires careful communication of predictive insights, stakeholder training in interpreting outputs, and clear governance structures guiding model use. AI tools in education must function as decision-support systems rather than automated decision-makers.

6. Methodological and operational considerations and limitations

Several considerations temper the interpretation of these findings. Although the sample size was substantial, variability in LMS usage patterns and data completeness introduced noise that may have influenced model behavior. Hyperparameter tuning was applied selectively, and while additional optimization might yield incremental improvements, such gains would require increased computational investment with uncertain practical benefits.

The inclusion of self-reported psychosocial measures introduces potential response biases associated with self-perception. While psychometric validation confirms reliability and construct validity, attitudinal indicators may fluctuate across semesters or instructional contexts. Additionally, the binary pass-fail classification simplifies academic performance, potentially masking gradations in achievement and developmental trajectories.

Generalizability represents a further limitation. The study was conducted within a single university context in Lesotho. Although this setting provides important representation within under-researched sub-Saharan environments, findings may not directly transfer to institutions with different digital infrastructures or demographic compositions. Moreover, the feature set did not incorporate fine-grained behavioral indicators such as time-on-task metrics, sequence-based interaction patterns, or social network structures, which may enhance future modelling efforts.

The fairness analysis relied primarily on statistical parity. While appropriate for assessing distributive balance in early-warning classification, additional metrics such as equalized odds, subgroup calibration, and intersectional analysis would provide a more comprehensive equity assessment. Finally, the study did not evaluate real-time implementation dynamics, including how educators interpret predictive dashboards or how students respond to AI-informed feedback. Understanding these human-AI interaction processes remain critical for responsible deployment.

7. Conclusion

This study demonstrates that machine learning can generate actionable insights into student performance when grounded in integrated behavioral and psychosocial data and evaluated through fairness-aware, interpretable analytic frameworks. Among the six classifiers examined, Random Forest, XGBoost, and Logistic Regression offered the most balanced combination of predictive accuracy, stability, and transparency. These models were particularly effective in identifying modifiable behavioral-motivational factors, including digital self-efficacy, discussion engagement, and feedback usefulness, which serve as meaningful instructional levers within LMS environments.

At the same time, the findings caution against uncritical adoption of highly complex models in digitally variable contexts. The instability and infrastructural sensitivity observed in neural network architectures illustrate how algorithmic systems may inadvertently amplify structural inequalities rather than isolate authentic learning signals. Model selection must therefore consider contextual constraints, data characteristics, and ethical implications alongside technical performance.

By integrating ensemble modelling with explainability and fairness diagnostics, the study contributes to the development of responsible AI-supported assessment systems. It offers a conceptual and methodological pathway for implementing early-warning tools that are transparent, contextually sensitive, and pedagogically actionable. Future research should extend fairness evaluation frameworks, examine longitudinal generalizability, and investigate how predictive feedback influences learner behavior and institutional decision-making. Advancing such work is essential to ensuring that AI-enhanced assessment systems meaningfully promote equitable learning opportunities across diverse higher education environments.

Data availability

The dataset used in this study is available from the corresponding author upon reasonable request.

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Ethical approval

All procedures performed in this study complied with institutional and international ethical standards. The research protocol was reviewed and approved by the Faculty Research Ethics Committee.

Consent to participate

Written informed consent was obtained from all participants prior to data collection.

Declaration of generative AI and AI-Assisted technologies in writing

The authors declare that they did not use Generative AI tools to develop this paper.

CRediT authorship contribution statement

Musa Adekunle Ayanwale: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Olasunkanmi Kehinde:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Mubarak O Mojoyinola:** Writing – review & editing, Writing – original draft, Supervision. **Chioma Ezech-Kehinde:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix - Instrument

Variable	Response Options	
Gender	1 = Male, 2 = Female	
Age Group	1 = 18–20 yrs, 2 = 21–23 yrs, 3 = 24–26 yrs	
Year of Study	1 = Year 1, 2 = Year 2, 3 = Year 3, 4 = Year 4	
Field of Study	1 = Education, 2 = Science & Technology, 3 = Social Sciences, 4 = Humanities	
Internet Reliability	1 = Very Poor, 2 = Poor, 3 = Fair, 4 = Good, 5 = Excellent	
Prior AI literacy: How would you rate your knowledge of AI tools used in education?	0 = No, 1 = Yes	
Consent: I agree that my LMS engagement data can be linked with my survey responses for research purposes.	0 = No, 1 = Yes	
LMS-Linked Behavioral Variables: Automatically extracted; not self-reported)		
Variable	Description	Scale / Coding
Attempt Frequency	Number of attempts made per LMS quiz/assignment (continuous).	0 – 10 (Scale data)
Discussion Engagement Level	Participation level in LMS forums (categorical).	0 = None, 1 = Low, 2 = Moderate, 3 = High, 4 = Very High
Feedback Response Behaviour	Whether student opened/acknowledged feedback within system logs.	0 = No, 1 = Yes
Survey-Based Psychosocial and Attitudinal Scales		
Self-Efficacy in Digital Learning (SEDL) (Ordinal 1–5)		
I can successfully complete online learning activities on my own.		
I am confident in navigating digital learning platforms.		
I can solve basic technical problems during online study.		
I can manage multiple online tasks effectively.		
I can guide peers on how to use the LMS.		
Feedback Usefulness Rating (FUR) (Ordinal 1–5)		
Feedback from my instructors helps me understand mistakes.		
Feedback guides how to improve my next submissions.		
Feedback arrives early enough to be useful.		
Feedback provided through the LMS is clear and actionable.		
I review feedback before submitting new work.		
AI Ethics Awareness (AIEA) (Ordinal 1–5)		
1. I understand basic ethical principles in AI use for education.		
2. I am aware that AI systems may reflect bias.		
3. I know how my data might be used by AI tools in assessment.		
4. I can recognise unfair AI decisions or outputs.		
5. I know where to report concerns about AI misuse in assessment.		
Trust in AI Assessment (TAIA) (Ordinal 1–5)		
I trust AI-generated assessment feedback to be accurate.		
I believe AI-based grading can be fair to students.		
I understand how AI systems in my LMS reach conclusions.		
I am comfortable acting on AI-generated feedback.		
I can challenge AI-generated results when I suspect an error.		

(continued on next page)

(continued)

Variable	Description	Coding
Assessment Score	Student's final assessment score derived from LMS gradebook	0 = Below 50 (Fail) 1 = 50 and Above (Pass)

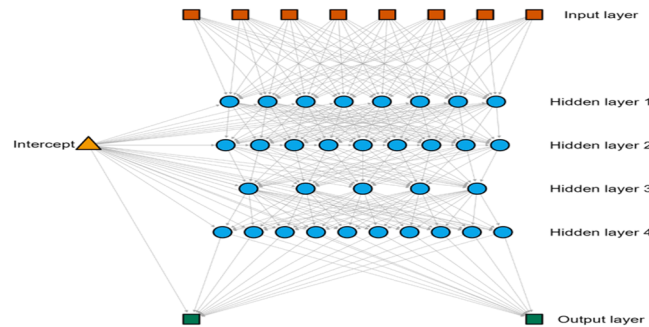


Fig. 1A. Multilayer perceptron.

Model performance evaluation

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}; Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times Precision}{Precision + Recall}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$$AUC = \int_0^1 TPR(FPR)d(FPR)$$

References

- [1] M.A. Ayanwale, Building trust in AI-powered assessment through explainable machine learning models, in: *Proceedings of the ACM Global on Computing Education Conference 2*, 2025, pp. 403–404. <https://dl.acm.org/doi/10.1145/3736251.3747313>.
- [2] R. Sharma, S.S. Shrivastava, A. Sharma, Predicting student performance using educational data mining and learning analytics technique, *J. Intell. Syst. Internet Things* 10 (2) (2023) 24–37, <https://doi.org/10.54216/JISIoT.100203>.
- [3] C.G. Nespereira, E. Elhariri, N. El-Bendary, A.F. Vilas, R.P.D Redondo, Machine learning-based classification approach for predicting students' performance in blended learning, in: *The 1st International Conference on Advanced Intelligent System and Informatics (AIS2015)*, November 28–30, 2015, Beni Suef, Egypt, Springer International Publishing, Cham, 2015, pp. 47–56, https://doi.org/10.1007/978-3-319-26690-9_5.
- [4] Maraza-Quispe, B., Damian Valderrama-Chauca, E., Henry Cari-Mogrovejo, L., & Milton Apaza-Huanca, J. (2021). Predictive model of student academic performance from lms data based on learning analytics. In *Proceedings of the 13th International Conference on Education Technology and Computers* (pp. 13–19). doi:10.1145/3498765.3498768.
- [5] S.B. Dias, S.J. Hadjileontiadou, J. Diniz, L.J. Hadjileontiadis, DeepLMS: a deep learning predictive model for supporting online learning in the Covid-19 era, *Sci. Rep.* 10 (1) (2020) 19888, <https://doi.org/10.1038/s41598-020-76740-9>.
- [6] A. Hernández-Blanco, B. Herrera-Flores, D. Tomás, B. Navarro-Colorado, A systematic review of deep learning approaches to educational data mining, *Complexity*. 2019 (1) (2019) 1306039, <https://doi.org/10.1155/2019/1306039>.
- [7] B. Sameephet, W. Chansanam, M. Rakhshandehroo, C. Srisawat, K. Nilubol, A. Jannok, K. Poonpon, A comparative analysis of machine learning models for predicting EFL student language performance in Smart learning environments, *Emerg. Sci. J.* 9 (2) (2025) 615–639, <https://doi.org/10.28991/ESJ-2025-09-02-07>.
- [8] D. Carless, D. Boud, The development of student feedback literacy: enabling uptake of feedback, *Assess. Eval. Higher Educ.* 43 (8) (2018) 1315–1325, <https://doi.org/10.1080/02602938.2018.1463354>.
- [9] R. Guidotti, A. Monreale, S. Ruggieri, F. Turini, F. Giannotti, D. Pedreschi, A survey of methods for explaining black box models, *ACM. Comput. Surv.* 51 (5) (2019) 1–42, <https://doi.org/10.1145/3236009>.
- [10] H. He, E.A. Garcia, Learning from imbalanced data, *IEEE Trans. Knowl. Data Eng.* 21 (9) (2009) 1263–1284, <https://doi.org/10.1109/TKDE.2008.239>.
- [11] S.M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, *Adv. Neural Inf. Process. Syst.* 30 (2017) 4765–4774. <https://dl.acm.org/doi/10.5555/3295222.3295230>.
- [12] R.S. Baker, T. Martin, L.M. Rossi, Educational data mining and learning analytics, *The Wiley Handb. Cognit. Assess.: Frameworks, Methodol., Appl.* (2016) 379–396, <https://doi.org/10.1002/9781118956588.ch16>.
- [13] Siemens, G., & Baker, R.S.D. (2012). Learning analytics and educational data mining: towards communication and collaboration. In *Proceedings of the 2nd International Conference on Learning Analytics and Knowledge* (pp. 252–254). doi:10.1145/2330601.2330661.
- [14] B.J. Zimmerman, Attaining self-regulation: a social cognitive perspective. *Handbook of Self-Regulation*, Academic press, 2000, pp. 13–39, <https://doi.org/10.1016/B978-012109890-2/50031-7>.
- [15] Y.J. Joo, K.Y. Lim, J. Kim, Locus of control, self-efficacy, and task value as predictors of learning outcome in an online university context, *Comput. Educ.* 62 (2013) 149–158, <https://doi.org/10.1016/j.compedu.2012.10.027>.
- [16] D. Nicol, D. Macfarlane-Dick, Formative assessment and self-regulated learning: a model and seven principles of good feedback practice, *Stud. High. Educ.* 31 (2006) 199–218, <https://doi.org/10.1080/03075070600572090>.
- [17] L. Floridi, J. Cowls, A unified framework of five principles for AI in society, *Harv. Data Sci. Rev.* 1 (2019) 2–15, <https://doi.org/10.1162/99608f92.8cd550d1>.
- [18] D. Long, B. Magerko, What is AI literacy? Competencies and design considerations, in: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, Association for Computing Machinery, 2020, pp. 1–16, <https://doi.org/10.1145/3313831.3376727>.
- [19] J.F. Hair, G.T.M. Hult, C.M. Ringle, M. Sarstedt, *Partial Least Squares Structural Equation Modeling (PLS-SEM)*, 3rd ed., Sage, 2021 <https://doi.org/10.1007/978-3-030-80519-7>.
- [20] Z. Swiecki, H. Khosravi, G. Chen, R. Martinez-Maldonado, J.M. Lodge, S. Milligan, N. Selwyn, D. Gašević, Assessment in the age of artificial intelligence, *Comput.*

- Educ.: Artif. Intell. 3 (2022) 100075, <https://doi.org/10.1016/j.caeai.2022.100075>.
- [21] L. Breiman, Random forests, Mach. Learn. 45 (1) (2001) 5–32, <https://doi.org/10.1023/A:1010933404324>.
- [22] C. Fornell, D.F. Larcker, Evaluating structural equation models with unobservable variables and measurement error, J. Market. Res. 18 (1) (1981) 39–50, <https://doi.org/10.2307/3151312>.
- [23] N. Kock, Common method bias in PLS-SEM: a full collinearity assessment approach, Int. J. E-Collabor. (IJEC) 11 (4) (2015) 1–10, <https://doi.org/10.4018/ijec.2015100101>.
- [24] R. Luckin, W. Holmes, M. Griffiths, L.B. Forcier, Intelligence unleashed: An argument For AI in Education, Pearson, 2016. https://www.researchgate.net/publication/299561597_Intelligence_Unleashed_An_argument_for_AI_in_Education.
- [25] Holstein, K., Wortman Vaughan, J., Daumé III, H., Dudik, M., & Wallach, H. (2019). Improving fairness in machine learning systems: what do industry practitioners need?. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (pp. 1–16). doi:10.1145/3290605.3300830.